

PROTEINLOOP tank

Off-grid AI for living protein systems

Gemma 4 | DECT NR+ | Deterministic safety

Simulate water emergency



AGENTIC AI CONTROL

Gemma configured

Live recovery mission

Mission

Recover water quality

Recover water quality

Reduce ammonia and restore dissolved oxygen while protecting fish and prawn survival.



5-agent team



Safety verifier



Human approval



Run agent team

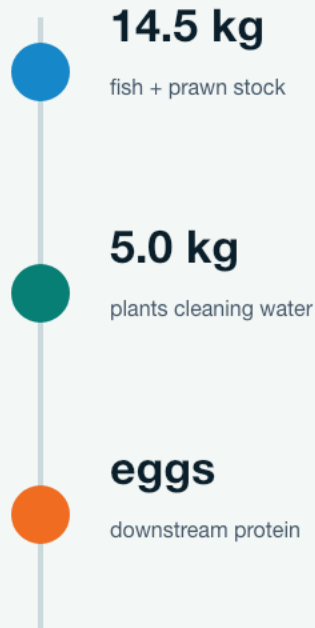
ONE LOOP, SHARED RISK

A chemistry failure does not stop at the vegetables.

Fish and prawns breathe the water. Plants clean it. Duckweed stores feed. Eggs depend on what survives upstream.

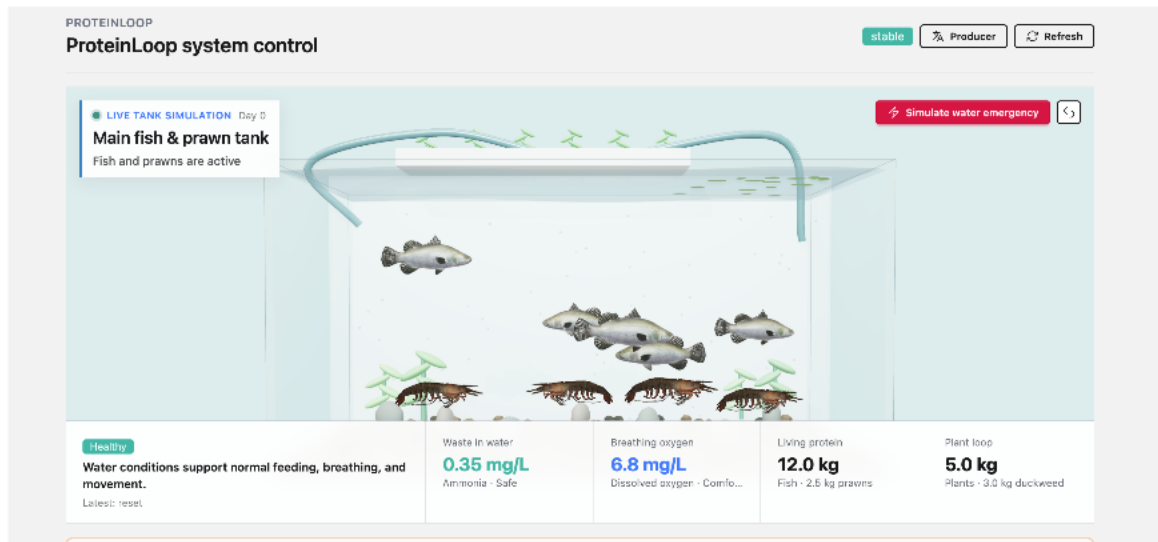
One ammonia spike

can threaten every protein output in the loop.



A living system people can understand in seconds.

The deployed operator view connects animal behavior, plain-language chemistry, biomass, and action in one scene.



SEE THE ANIMALS

UNDERSTAND THE WATER

ACT WITH EVIDENCE

The recovery is visible while it happens.

1 Observe

2 4 specialists

3 Supervisor

4 Verify

AGENTIC AI CONTROL

Live recovery mission

5 Measure

Mission

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Structured events and tool outcomes are visible; private model chain-of-thought is not presented as evidence.

Needs attention

Waste in water

0.75

Breathing oxygen

0.25

Living protein

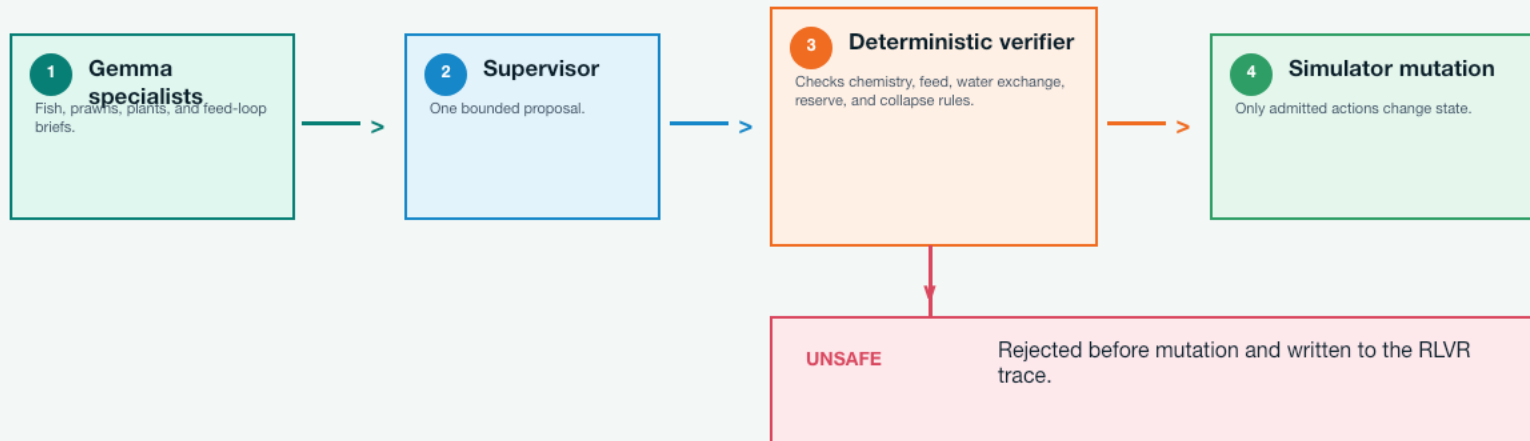
100%

Plant loop

5.00%

The model can recommend. It cannot mutate.

Authority moves left to right. Every connector ends at an explicit decision boundary.



Safety is demonstrated, not narrated.

The simulator is simultaneously the product demo, the policy evaluator, and the RLVR source of truth.

220

Python tests

simulator, verifier, API, evidence

141

Phoenix tests

Sagents, LiveView, producer control

+463

reward delta

verified policy versus naive average

100%

collapse avoidance

fixed baseline scenarios recovered

ONE-CLICK JUDGE PROOF

Inject emergency -> block unsafe proposal -> admit safe recovery -> inspect measured chemistry

Keep the food-control loop local.

Three continuity layers separate transport, intelligence, and power claims.



NO WI-FI

DECT NR+ carries the tank-to-gateway hop.

NO CLOUD

Self-hosted Gemma and deterministic rules remain local.

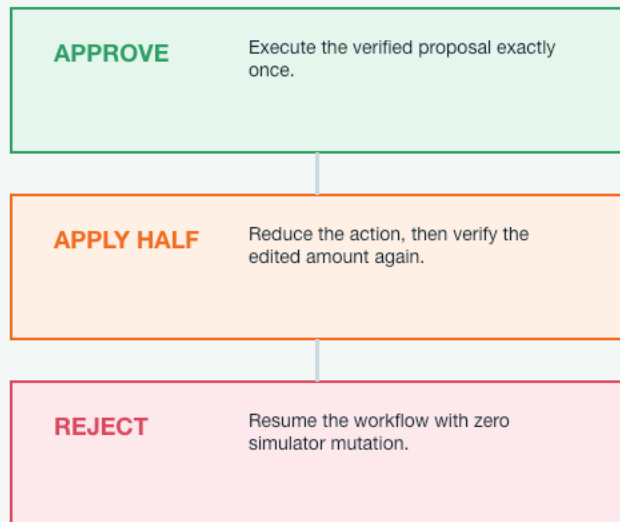
NO GRID

Solar + battery is the next measured autonomy proof.

Risky actions stop for a human decision.

The producer sees the proposed change, safety result, and current chemistry before choosing what happens next.

No irreversible action runs silently.



Verifier feedback turns 10% first-pass safety into 100%.

Exact verifier failures return as bounded feedback; every revision is parsed and verified again.

CAPTURED AMD RUNTIME

Gemma 4 E2B

- PyTorch 2.10 · ROCm 7.2
- vLLM 0.20.2 · gfx1100
- 47.98 GiB AMD GPU memory

20-EMERGENCY REPAIR

18 repairs · zero fallback

- 2/20 first answers safe
- 17 repaired once · 1 twice
- 20/20 model-safe · no weight update

MEASURED EXECUTION

139 requests

- 60,385 observed tokens
- 99.793 completion tokens/s
- p50 latency · 655.522 ms

420.648 kg aggregate scenario biomass · +221.7244 mean reward vs naive

Protect protein production where connectivity cannot be assumed.

A verifier-gated, producer-controlled platform for rural farms, cooperatives, and food-security programs.

PRIVATE FIELD LINK	Two physical nRF9151 radios
LOCAL INTELLIGENCE	Self-hosted Gemma 4
EXECUTABLE SAFETY	Deterministic verifier + RLVR

proteinloop.dev-vb.lat | Best Unicorn

github.com/Anarpego/ProteinLoop

